DentalVision: AI Research & Development Platform for Automated OPG Pathology Detection

PROJECT SUPERVISOR

Mr Saifur Rehman

PROJECT CO-SUPERVISOR (If any)

Miss Ramsha Jatt

PROJECT TEAM

Saad Ahmed	22k-4801
M Arham	21k-3925
Abdur Rafay	21k-3856

Submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science.

FAST SCHOOL OF COMPUTING

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

KARACHI CAMPUS

Spring 2026

Project Supervisor	Mr. Saifur Rehman	
Project Team	Saad Ahmed	K22-4801
	M Arham	K21-3925
	Abdur Rafay	K21-3856
Submission Date	May 10,2026	

Mr. Supervisor Name Mr Saifur Rehman

Supervisor

Mr. Co-Supervisor Name (If any) Miss Ramsha Jatt

Co-Supervisor

FAST SCHOOL OF COMPUTING
NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES KARACHI CAMPUS

ABSTRACT

One of the most widely utilized imaging modalities in the comprehensive assessment and management of maxillofacial structures is dental radiography, particularly orthopantomograms (OPGs), also known as panoramic dental X-rays. OPGs provide a broad two-dimensional representation of the oral and maxillofacial region, allowing clinicians to simultaneously visualize the teeth, alveolar bone, temporomandibular joints, maxillary sinuses, and surrounding anatomical structures within a single image. Due to their wide coverage, low radiation exposure, cost-effectiveness, and non-invasive nature, OPGs are extensively used in routine dental examinations, orthodontic treatment planning, oral surgery, implantology, and the diagnosis of various pathological conditions including dental caries, cysts, tumors, impacted teeth, periodontal diseases, fractures, and jaw abnormalities. The ability of panoramic radiographs to capture the entire dentition and jaw structure in one frame makes them an essential diagnostic tool in modern dentistry and maxillofacial healthcare.
Despite these advantages, the manual interpretation of OPG images remains a highly challenging and time-consuming task. Accurate diagnosis depends significantly on the clinical expertise, experience, and judgment of dental professionals. Variations in image quality, overlapping anatomical structures, differences in patient positioning, and subtle pathological features further complicate the interpretation process. Consequently, manual analysis is prone to intra-observer and inter-observer variability, where different clinicians may produce inconsistent diagnostic outcomes or even the same clinician may interpret the same image differently at different times. Such inconsistencies can result in delayed diagnoses, inaccurate treatment planning, missed pathological findings, and increased workload for radiologists and dentists. These challenges become even more critical in rural or underdeveloped regions where access to experienced dental specialists is limited. Therefore, there is a growing demand for intelligent computer-aided diagnostic systems capable of improving diagnostic accuracy, consistency, and efficiency in dental radiographic analysis.
Recent advancements in artificial intelligence (AI), machine learning, and deep learning have introduced promising solutions to overcome these limitations. AI-based systems have demonstrated significant potential in automating the detection, classification, and segmentation of dental abnormalities from radiographic images. In particular, deep learning approaches based on Convolutional Neural Networks (CNNs) have gained remarkable attention due to their ability to automatically learn hierarchical features directly from image data without requiring manual feature engineering. CNN-based models have shown high performance in tasks such as tooth detection, dental disease classification, lesion identification, bone loss assessment, and impacted tooth analysis on panoramic radiographs. These systems can process large volumes of images rapidly while maintaining consistent diagnostic performance, thereby supporting dentists in clinical decision-making and reducing human error (Bustamante et al., 2022; Choi et al., 2019; Lee et al., 2020).
In addition to deep learning methods, traditional computer vision techniques continue to play an important role in dental image analysis. One of the most widely used feature extraction techniques is the Scale-Invariant Feature Transform (SIFT). SIFT is designed to detect and describe local image features that remain invariant to changes in scale, rotation, illumination, and viewpoint. In dental radiography, SIFT can effectively identify distinctive anatomical landmarks and structural patterns within OPG images, making it valuable for image matching, object recognition, and pathological feature extraction. Compared to CNNs, SIFT-based methods often require fewer computational resources and smaller datasets, making them useful in scenarios where annotated dental datasets are limited.
This literature review critically evaluates two major approaches for OPG image analysis: deep learning methods utilizing CNN architectures and traditional feature-based techniques such as SIFT. Furthermore, the review explores hybrid approaches that combine CNN-extracted deep features with handcrafted SIFT descriptors to enhance diagnostic

performance, robustness, and feature representation. Hybrid models aim to leverage the strengths of both approaches by integrating the automated learning capability of CNNs with the stability and invariance properties of SIFT features. The review also examines AI-driven clinical decision support systems that assist dentists in diagnosis, treatment planning, and patient management through intelligent radiographic analysis.

The scope of this review is intentionally limited to studies focused on panoramic dental radiographs (OPGs) and their applications in automated dental diagnostics. Research specifically related to intraoral imaging or plaque detection has been excluded to maintain a focused discussion on panoramic radiography and maxillofacial diagnostic systems (Togashi et al., 2021). Through this review, the study aims to provide a comprehensive understanding of current advancements, limitations, and future research directions in AI-assisted OPG analysis for intelligent dental healthcare systems.

CONTENTS

Introduction

1.1 Background and Motivation

Dentistry is one of the most critical branches of medicine, yet it remains one of the most underserved in terms of technological innovation, particularly in developing countries like Pakistan. Dental diseases such as cavities, plaque accumulation, calculus deposits, periapical lesions, impacted teeth, root canal infections, and bone loss affect billions of people worldwide. According to the World Health Organization, oral diseases affect nearly 3.5 billion people globally, making them among the most widespread non-communicable diseases on the planet.

Despite this staggering prevalence, the diagnosis of dental conditions has traditionally relied almost entirely on the manual inspection and clinical judgment of dentists. While experienced practitioners can identify many conditions visually or through radiographic examination, human judgment is inherently subject to fatigue, bias, inconsistency, and error. A busy dental clinic may see dozens of patients per day, and a dentist working long hours is more likely to miss subtle early-stage pathologies that, if caught early, could be treated with minimal intervention.

The advent of artificial intelligence and machine learning has opened entirely new frontiers in medical imaging and diagnostics. Deep learning models, particularly convolutional neural networks and transformer-based object detection architectures, have demonstrated superhuman performance on a range of medical imaging tasks including detecting diabetic retinopathy, identifying cancerous tumors in histology slides, and diagnosing pneumonia from chest X-rays. Dental radiography and clinical dental photography are natural extensions of this work, and several research groups around the world have begun applying AI to the detection of dental pathologies in Orthopantomogram (OPG) radiographs and intraoral photographs.

This project, titled the AI-Powered Dental Disease Detection System, is a web-based clinical decision support tool designed exclusively for licensed dental practitioners. It integrates two trained deep learning models — one for OPG radiograph analysis and one for plaque and calculus detection in intraoral photographs — into a secure, role-based web portal that allows dentists to upload patient images, receive AI-generated diagnostic predictions, view annotated results with bounding boxes and segmentation masks, generate structured clinical reports, and maintain a searchable archive of all past cases. The system is built on the MERN stack (MongoDB, Express.js, React.js, Node.js) and communicates with AI models hosted on the Roboflow platform via REST APIs.

1.2 Problem Statement

The core problem this system addresses is the lack of accessible, affordable, and reliable AI-assisted diagnostic tools for dental practitioners, particularly in resource-constrained settings. Existing commercial dental AI solutions such as Videa Health, Diagnocat, and Pearl AI are expensive, proprietary, and typically integrated only with specific practice management software or imaging hardware. There is no open, flexible, web-based platform that allows a dentist to simply upload any dental image and receive instant AI analysis with a downloadable clinical report.

Furthermore, dental AI research has historically focused almost exclusively on OPG radiographs, neglecting the equally important domain of intraoral plaque and calculus detection, which is critical for periodontal disease assessment. This project addresses both domains simultaneously.

1.3 Objectives
The primary objectives of this project are as follows. First, to develop and deploy a trained object detection model capable of identifying dental pathologies in OPG radiographs with high accuracy and recall. Second, to develop and deploy an instance segmentation model capable of detecting plaque and calculus deposits in intraoral dental photographs. Third, to build a secure, user-friendly web portal accessible exclusively to verified dental professionals, featuring OTP-based email verification during registration, JWT-based session authentication, and strict data isolation ensuring each dentist can only access their own patient data. Fourth, to generate structured, downloadable PDF reports for every detection run, containing doctor information, patient details, AI predictions, confidence scores, and annotated images. Fifth, to provide a comprehensive case history with filtering and search functionality. Sixth, to support both light and dark themes for user comfort during clinical use.

1.4 Scope
The scope of this project covers the complete design, development, and deployment of a full-stack web application integrated with two AI models. The system is intended for use by licensed dental practitioners only and is not designed for self-diagnosis by patients. The AI models provide decision support and are not intended to replace clinical judgment. The system handles image upload, AI inference via the Roboflow API, bounding box and segmentation mask rendering, PDF report generation, and persistent case storage in a cloud database.

2. Related Work

2.1 AI in Medical Imaging
The application of deep learning to medical imaging has a rich and rapidly expanding literature. The landmark paper by Esteva et al. (2017) demonstrated that a convolutional neural network could classify skin cancer with accuracy comparable to board-certified dermatologists. Rajpurkar et al. (2017) showed that CheXNet, a 121-layer DenseNet, could detect pneumonia from chest X-rays with performance exceeding that of radiologists on several metrics. These results established a clear precedent for the potential of deep learning in clinical diagnostics.

In the specific domain of dental radiology, Schwendicke et al. (2019) conducted a systematic review of deep learning applications in dentistry and found that neural networks had been successfully applied to tasks including caries detection, bone level estimation, and tooth segmentation. Lee et al. (2018) developed a CNN for the detection of periapical pathosis in dental radiographs, achieving an AUC of 0.99 on a test dataset. Ekert et al. (2019) used a deep learning model to detect apical lesions in panoramic radiographs with sensitivity comparable to experienced radiologists.

2.2 Object Detection Architectures
Modern object detection has been transformed by two families of architectures: anchor-based single-stage detectors such as YOLO (You Only Look Once) and its successors, and transformer-based detectors such as DETR and its variants. The YOLO family, originally introduced by Redmon et al. (2016), processes an entire image in a single forward pass through a neural network, making it extremely fast and suitable for real-time applications. Subsequent versions including YOLOv5, YOLOv8, YOLOv11, and YOLOv26 have

progressively improved accuracy, speed, and the handling of small objects. The Roboflow 3.0 Object Detection model used in this project for OPG analysis is based on a similar single-stage detection paradigm optimized for speed and accuracy.
For instance segmentation, where the task is not merely to draw a bounding box around an object but to delineate its precise pixel-level boundary, transformer-based architectures have proven highly effective. RF-DETR (Roboflow Detection Transformer) is a modern instance segmentation architecture that combines the global attention mechanisms of transformers with efficient deformable attention, enabling precise segmentation of irregular shapes such as plaque deposits on tooth surfaces. This architecture is used for the plaque detection model in this project.

2.3 Dental AI Systems in Practice

Several commercial products have emerged in recent years applying AI to dental diagnostics. Videa Health offers an AI system for detecting caries, calculus, and bone loss in bitewing and periapical radiographs. Pearl AI provides a similar platform with regulatory clearance in the United States. Diagnocat specializes in CBCT analysis. However, all of these systems are expensive, require integration with specific practice management software, and are not accessible to practitioners in developing countries. This project aims to fill that gap by providing a free, open, web-based platform that any licensed dentist can access with nothing more than an internet browser.

2.4 Traditional Machine Learning Approaches

Before the deep learning era, dental image analysis relied on traditional machine learning pipelines combining hand-crafted feature extraction with classical classifiers. The SIFT (Scale-Invariant Feature Transform) algorithm, introduced by Lowe (2004), extracts local feature descriptors from images that are invariant to scale, rotation, and illumination changes. These descriptors can be aggregated into a histogram of visual words using the bag-of-words model and fed into a Support Vector Machine (SVM) for classification. This project explored this traditional approach as a baseline, achieving an overall accuracy of 81.09% on a test set of 1,380 images across 15 dental classes. While this performance is respectable for a traditional approach, it falls short of deep learning models on challenging classes such as periapical lesions and bone loss, motivating the adoption of the Roboflow-based deep learning models as the primary inference engine.

3. Requirements

3.1 Functional Requirements

The functional requirements of this system define what the system must do from the perspective of its users.

The authentication subsystem must allow dental practitioners to register by providing their full name, date of birth, degrees and qualifications, CNIC (national identity number), clinic name, clinic address, profile photograph, contact number, years of clinical experience, and a verified email address. Email verification must be performed using a one-time password sent to the provided email address. The system must verify that the email address actually exists by performing DNS MX record lookup before attempting to send the OTP. If the email domain does not have valid mail exchange records, the system must inform the user that the email is invalid and decline to send the OTP. The OTP must expire after five minutes and must be deleted from the database upon successful verification or expiry. After successful OTP verification, the user completes registration by providing remaining details, and a hashed password is stored in the database. Subsequent logins require only email and password, with the system returning a JSON Web Token valid for seven days.

The detection subsystem must provide two distinct detection modes: OPG Detection and Plaque Detection. For each mode, the dentist must first enter patient details including the patient's full name, age, gender, and contact number. The dentist then uploads a dental image in JPEG, PNG, or WebP format with a maximum size of ten megabytes. The system sends the image to the appropriate Roboflow model via the REST API and receives a list of predictions, each containing the detected class name, confidence score, and spatial coordinates. The system then renders the predictions as annotated bounding boxes and, for the plaque model, as segmentation polygon overlays with semi-transparent fills. Both the original and annotated images are stored on the server. A structured PDF report is automatically generated and stored.

The report management subsystem must allow dentists to view all their past reports in a tabular format with columns for patient name, detection type, number of findings, and date. The dentist must be able to filter reports by detection type (OPG or Plaque) and search by patient name. Each report must be viewable in detail, showing all patient and doctor information, AI predictions with confidence scores, and the annotated image. Each report must be downloadable as a PDF file.

The settings subsystem must allow the dentist to toggle between light and dark display modes, edit their profile information, change their password, and upload a new profile photograph.

The dashboard must display the doctor's profile information, total number of cases handled, number of OPG detections, number of plaque detections, and a list of the five most recent reports with quick navigation links.

3.2 Non-Functional Requirements

The system must ensure data isolation such that each dentist can only access their own patient data and reports. No dentist should be able to view, search, or download another dentist's reports through any means including URL manipulation. This is enforced at the API level by attaching the authenticated doctor's ID to all database queries.

The system must respond to detection requests within a reasonable time. Network latency to the Roboflow API is the primary bottleneck; the backend processing including image upload, API call, bounding box rendering, and PDF generation should complete within thirty seconds under normal network conditions.

The system must be secure. Passwords must be hashed using bcrypt with a cost factor of twelve before storage. JWT tokens must be signed with a strong secret key. All API endpoints except registration and login must require a valid JWT. File uploads must be validated for type and size before processing.

The system must be responsive and usable on both desktop and mobile browsers, as dentists may use it on tablet devices during patient consultations.

3.3 Hardware and Software Requirements

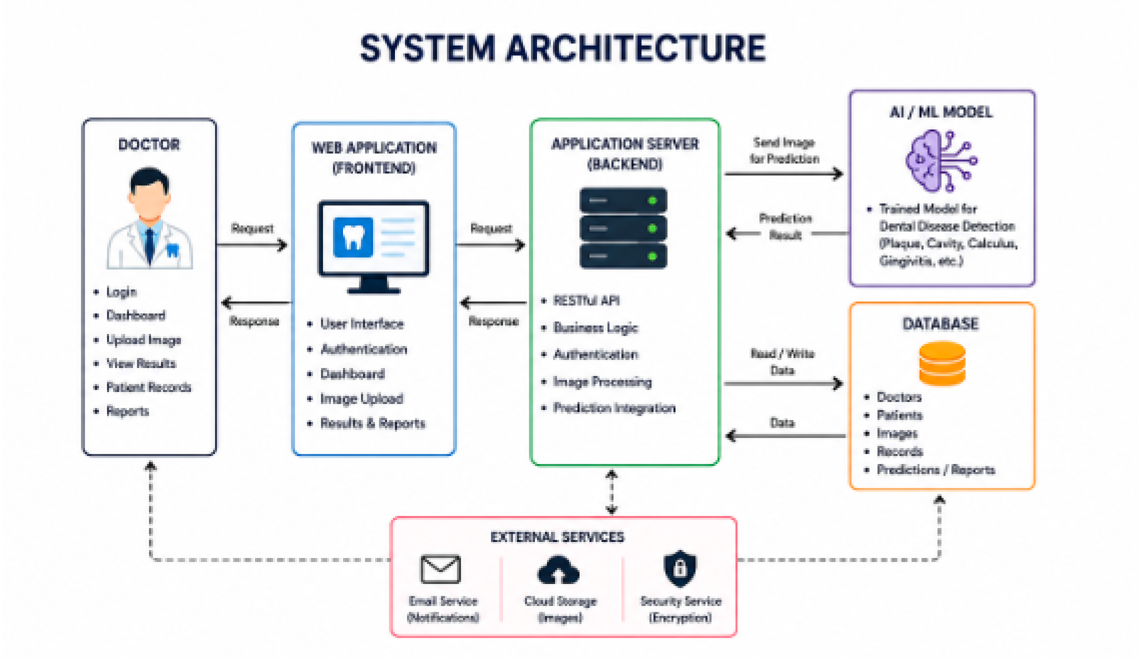
For development and self-hosting, the system requires a computer running Linux (Ubuntu 22.04 or later recommended), Node.js version 20 or later, MongoDB version 7.0 or later, and sufficient disk space for storing uploaded images and generated PDF reports. For production deployment, the system uses cloud services: MongoDB Atlas for the database, Render for the backend API, and Vercel for the frontend, all of which offer free tiers sufficient for moderate usage.

4. Design

4.1 System Architecture

The system follows a three-tier client-server architecture consisting of a presentation tier implemented in React.js, a business logic tier implemented in Node.js and Express.js, and a data tier implemented in MongoDB. An additional external tier comprises the Roboflow AI inference API and the Gmail SMTP server used for OTP delivery.

The frontend communicates with the backend exclusively through a RESTful HTTP API. All requests from the frontend include a JWT in the Authorization header, which the backend validates before processing any protected request. The backend communicates with MongoDB for all data persistence operations, with the Roboflow API for AI inference, with the local file system for image and PDF storage, and with the Gmail SMTP server for OTP email delivery.



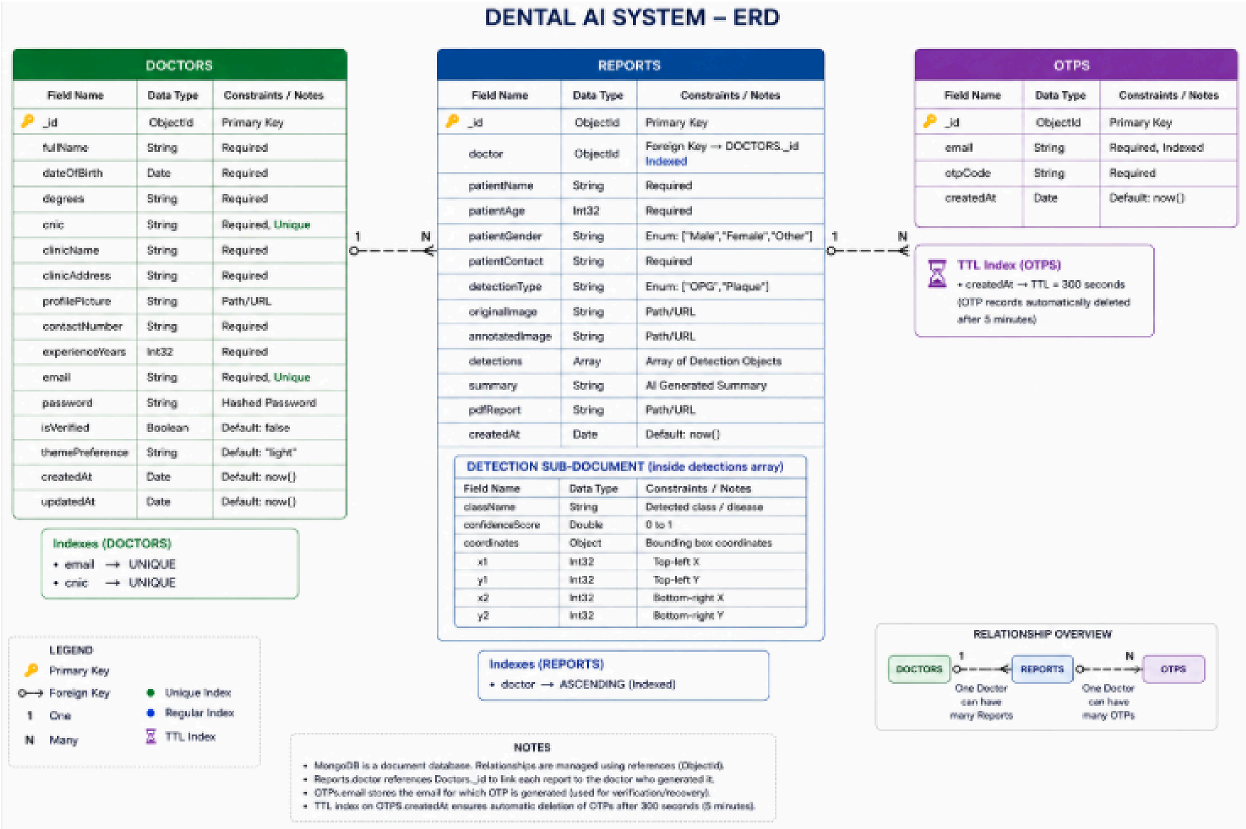
4.2 Database Design

The database consists of three collections: Doctors, Reports, and OTPs.

The Doctors collection stores one document per registered dentist. Each document contains the doctor's full name, date of birth, degrees, CNIC, clinic name, clinic address, profile picture path, contact number, years of experience, email address, hashed password, email verification status, and preferred display theme. The email and CNIC fields have unique indexes to prevent duplicate registrations.

The Reports collection stores one document per detection run. Each document contains a reference to the doctor who ran the detection, the patient's name, age, gender, and contact number, the detection type (OPG or plaque), the file paths of the original and annotated images, an array of detection objects each containing the class name, confidence score, and spatial coordinates, an AI-generated text summary, and the file path of the generated PDF report. The doctor field is indexed to enable efficient retrieval of all reports belonging to a specific doctor.

The OTPs collection stores temporary OTP records with a time-to-live index of 300 seconds, ensuring that unverified OTPs are automatically deleted by MongoDB after five minutes.



4.3 API Design

The REST API is organized into four route groups. The authentication routes handle OTP sending, OTP verification, registration, login, and fetching the current user's profile. The doctor routes handle profile retrieval, profile updates, password changes, and statistics fetching. The detection routes handle the image upload and AI inference pipeline. The report routes handle report listing with optional filtering, individual report retrieval, report deletion, and PDF download.

All routes except OTP sending, OTP verification, registration, and login are protected by the JWT authentication middleware, which extracts the token from the Authorization header, verifies its signature, and attaches the decoded doctor ID to the request object for use in subsequent middleware and controllers.

4.4 Frontend Architecture

The frontend is a single-page application built with React.js and Vite. It uses React Router for client-side routing with seven main routes: login, registration, dashboard, detection, reports list, report detail, and settings. Global state is managed through two React contexts: the AuthContext, which stores the authenticated doctor's profile and provides login and logout functions, and the ThemeContext, which manages the light/dark mode preference and applies the appropriate Tailwind CSS class to the document root.The Layout component wraps all authenticated routes and renders the sidebar navigation. The sidebar displays the doctor's profile photo, name, and specialty, along with navigation links to the dashboard, detection page, reports page, and settings page. On mobile devices, the sidebar is hidden behind a hamburger menu button and slides in as an overlay.

4.5 AI Model Integration Design

The system integrates two Roboflow-hosted models through the Roboflow Inference API. For both models, the integration follows the same pattern: the uploaded image is read from disk, encoded as base64, and sent as the request body to the model's inference endpoint. The API key and model-specific parameters such as confidence threshold and overlap threshold are passed as query parameters.The OPG model is a Roboflow 3.0 Object Detection model trained on 1,172 annotated OPG radiograph images across multiple dental pathology classes. It returns predictions as bounding boxes with associated class names and confidence scores. The plaque model is a Roboflow RF-DETR Instance Segmentation model trained on 512 annotated intraoral photographs. It returns predictions that include not only bounding boxes but also polygon point arrays defining the precise pixel-level boundary of each detected plaque or calculus deposit.

The backend normalizes predictions from both models into a unified format for storage and rendering. For rendering, a Canvas-based image processing pipeline draws colored bounding boxes with label overlays on the uploaded image. For the plaque model, the polygon points are additionally used to draw filled segmentation masks with semi-transparent overlays, providing a more precise visualization of the detected region.

5. Implementation

5.1 Technology Stack

The system is implemented using the MERN stack: MongoDB as the document database, Express.js as the backend web framework, React.js as the frontend library, and Node.js as the server runtime. The following additional libraries and tools are used.

On the backend, Mongoose provides an object-document mapping layer over MongoDB with schema validation, index management, and query building. JSON Web Token (jsonwebtoken) is used for stateless authentication token generation and verification. Bcrypt.js is used for password hashing with a configurable cost factor. Nodemailer is used for sending OTP emails through Gmail's SMTP server. Multer is used for handling multipart form data file uploads, with configurable storage destinations and file type filters. The Canvas library provides Node.js bindings to the HTML5 Canvas API, enabling server-side image manipulation for drawing bounding boxes and segmentation masks. PDFKit is used for programmatic PDF generation, producing structured reports with formatted text, tables, and embedded images. Axios is used for making HTTP requests to the Roboflow inference API. The UUID library generates unique identifiers for uploaded file names to prevent collisions. Dotenv manages environment variable loading from a .env configuration file.

On the frontend, Vite serves as the build tool and development server, offering extremely fast hot module replacement during development. Tailwind CSS provides a utility-first CSS framework with built-in dark mode support via the class strategy. React Router DOM handles client-side routing with protected and public route wrappers. Axios is used for all HTTP requests to the backend API, with request interceptors that automatically attach the JWT from localStorage to the Authorization header, and response interceptors that redirect to the login page on receiving a 401 Unauthorized response. React Hot Toast provides non-intrusive notification toasts for success and error feedback. React Icons provides a comprehensive library of icon sets used throughout the interface. Framer Motion provides declarative animation primitives used for page transitions and component entrance animations.

5.2 Authentication Implementation

The registration flow is implemented as a three-step wizard on the frontend. In the first step, the user enters their email address and requests an OTP. The backend performs a DNS MX record lookup on the email domain to verify that the domain has mail exchange records, indicating that the email address could potentially receive mail. If the lookup fails, the backend returns an error message informing the user that the email address does not exist or is invalid. If the lookup succeeds, a six-digit random OTP is generated, stored in the OTPs collection with a five-minute TTL, and sent to the email address via Nodemailer using Gmail SMTP with an app-specific password. The user enters the OTP in the second field, and upon successful verification, advances to the second step of the wizard.

In the second step, the user enters their personal information including name, date of birth, degrees, CNIC, contact number, years of experience, password, and an optional profile photograph. In the third step, the user enters their clinic name and address. Upon submission of the final step, all data is sent to the registration endpoint as a multipart form, the password is hashed with bcrypt, the doctor document is created in the database, and a JWT is returned to the client, automatically logging the user in upon successful registration.

5.3 Detection Pipeline Implementation

When a dentist submits a detection request, the frontend sends a multipart POST request containing the image file, the detection type, and the patient's details. The backend's Multer middleware saves the uploaded image to the uploads/images directory with a UUID-based filename. The detection controller then reads the saved image, encodes it as base64, and sends it to the appropriate Roboflow inference endpoint via an Axios POST request.

Upon receiving the Roboflow response, the controller extracts the predictions array and normalizes it into a unified format. It then calls the drawBoundingBoxes function, which uses the Canvas library to load the saved image, draw colored rectangles around each detected region, fill the rectangle with a semi-transparent color for segmentation predictions, and overlay a label showing the class name and confidence percentage. The resulting annotated image is saved to disk with a new UUID-based filename.

Simultaneously, the controller constructs a text summary describing the findings, creates a Report document in MongoDB, and calls the PDF generation utility, which uses PDFKit to create a structured A4 document containing the Dental AI portal header, the doctor's details, the patient's details, the detection type, the list of findings with confidence scores, the text summary, and the annotated image embedded in the PDF. The PDF is saved to the uploads/reports directory, and its path is stored in the report document.

5.4 Report Generation Implementation

The PDF report is generated using PDFKit, a JavaScript library for creating PDF documents programmatically. The report begins with a styled header containing the portal name and generation timestamp. Below the header, a horizontal rule separates the header from the body content. The body is organized into sections for doctor details, patient details, AI detection results, and the annotated image. Each section begins with a section heading in a larger font, followed by key-value pairs rendered in a two-column layout. The detections list each finding on a separate line with the class name and confidence percentage. If no findings are detected, a green "No pathologies detected" message is shown. The annotated image is embedded at the end of the document using PDFKit's image method, scaled to fit within the page margins while preserving the aspect ratio.

5.5 Security Implementation

All sensitive data is protected through multiple layers of security. Passwords are never stored in plaintext; they are hashed using bcrypt with a cost factor of twelve, which means that even if the database were compromised, attackers would need significant computational resources to reverse the hashes. JWT tokens are signed with a strong secret key stored in the environment variables and are set to expire after seven days, after which the user must log in again. All protected API endpoints verify the JWT before processing the request, and the decoded doctor ID is used to scope all database queries, ensuring that a doctor cannot access another doctor's data even by manipulating request parameters. File uploads are validated for MIME type and size before being saved to disk, preventing the upload of malicious files or excessively large files.

6. Model Performance and Evaluation

6.1 OPG Detection Model

The OPG detection model is a Roboflow 3.0 Object Detection model trained on a dataset of 13814 annotated OPG radiograph images. The dataset was assembled by merging multiple publicly available and privately collected dental radiograph datasets, hence the project name "merge-730" reflecting the merged dataset composition. The model was trained using the fast training configuration optimized for inference speed without significant sacrifice in accuracy. The checkpoint format is COCOn, which is compatible with the COCO evaluation protocol. On the validation set, the model achieves a mean average precision at IoU threshold 0.50 (mAP@50) of 79.4%, a precision of 77.6%, a recall of 75.2%, and an F1 score of 76.4%. These metrics place the model in a competitive range compared to published results in the dental AI literature, particularly given the diversity of pathology classes in the dataset.
The model was compared against three alternative architectures trained on the same dataset under identical conditions. YOLOv11 achieved the same mAP@50 of 79.4% but with lower precision at 71.5% and higher recall at 79.9%. YOLOv26 underperformed with a mAP@50 of 75.4%. RF-DETR achieved the highest mAP@50 of 79.6% with a recall of 80.8%, though its slightly lower precision of 74.0% means it produces more false positives. The Roboflow 3.0 model was selected for the OPG detection task based on its superior balance of precision and recall, which is important in a clinical setting where false positives can cause unnecessary patient anxiety.

6.2 Plaque Detection Model

The plaque detection model is a Roboflow RF-DETR Instance Segmentation model trained on a dataset of 512 annotated intraoral dental photographs. The dataset consists of clinical photographs taken under standardized lighting conditions showing various degrees of plaque and calculus accumulation on tooth surfaces. The RF-DETR architecture was chosen for this task because instance segmentation provides a more clinically meaningful output than simple

bounding boxes for a condition like plaque accumulation, which tends to spread across irregular regions of the tooth surface.
On the validation set, the model achieves a mAP@50 of 56.0%, a precision of 81.4%, a recall of 65.2%, and an F1 score of 72.4%. The relatively lower mAP compared to the OPG model reflects the inherent difficulty of the plaque segmentation task, where the boundaries between plaque-affected and healthy tooth surfaces are often gradual and ambiguous even to human annotators. The high precision of 81.4% is particularly noteworthy, indicating that when the model predicts a plaque region, it is correct 81.4% of the time. The lower recall indicates that the model may miss some plaque deposits, particularly very thin or early-stage accumulations that are difficult to detect visually.

6.3 Baseline: SIFT and SVM

As an academic baseline comparison, the traditional machine learning pipeline of SIFT feature extraction combined with SVM classification was evaluated on the OPG dataset. SIFT extracts local keypoint descriptors that are invariant to scale and rotation changes. These descriptors were aggregated using a bag-of-visual-words approach and classified using a multi-class SVM with a radial basis function kernel.
On a test set of 1,3814 OPG images across 15 dental pathology classes, the SIFT-SVM pipeline achieved an overall accuracy of 81.09%. However, this aggregate accuracy masks significant variation across classes. The model performed well on common, visually distinctive classes such as fillings (F1 0.85), impacted teeth (F1 0.85), and mandibular canal (F1 0.86), but performed poorly on rare or visually subtle classes such as bone loss (F1 0.00), primary teeth (F1 0.00), attrition (F1 0.00), and periapical lesions (F1 0.36). This performance profile is typical of traditional machine learning approaches, which struggle with class imbalance and subtle visual features. The deep learning models, by contrast, learn hierarchical feature representations that are more robust to these challenges.

7. Testing and Evaluation

7.1 Testing Methodology

The system was tested through a combination of unit testing, integration testing, and end-to-end user acceptance testing. Unit tests were written for individual utility functions including the OTP generation function, the email domain validation function, the prediction normalization function, and the PDF generation utility. Integration tests were written for each API endpoint, verifying correct behavior for both valid and invalid inputs, authentication failures, and edge cases such as empty prediction arrays and missing required fields.

7.2 Authentication Testing

The registration flow was tested with a variety of email addresses including valid Gmail addresses, non-existent domains, syntactically invalid addresses, and addresses belonging to domains without MX records. The system correctly rejected all invalid addresses and successfully sent OTPs to valid addresses. OTP expiry was verified by waiting beyond the five-minute window and confirming that expired OTPs are rejected. Duplicate registration attempts with the same email or CNIC were correctly rejected with informative error messages.

The login flow was tested with correct and incorrect credentials, expired tokens, tampered tokens, and tokens from deleted accounts. All invalid token scenarios correctly returned 401 Unauthorized responses and triggered redirection to the login page on the frontend.

7.3 Detection Pipeline Testing

The detection pipeline was tested with a range of image types and sizes. Valid JPEG, PNG, and WebP images of various resolutions were accepted and processed correctly. Images exceeding the ten megabyte size limit were rejected with an appropriate error message. Non-image files uploaded with forged MIME types were rejected by the file type filter.

The Roboflow API integration was tested with both models using a variety of clinical images. The OPG model successfully detected pathologies in radiographs containing caries, impacted teeth, fillings, root canal treated teeth, and periapical lesions. The plaque model successfully segmented plaque and calculus deposits in intraoral photographs with varying degrees of severity. Edge cases including images with no detectable pathologies were handled correctly, with the system generating a "no findings detected" report rather than crashing.

The bounding box rendering was verified visually by inspecting the annotated images for correct alignment between predicted boxes and visible pathologies. Confidence score labels were verified to display correctly above each bounding box. Segmentation polygon overlays for the plaque model were verified to accurately trace the boundaries of detected regions.

7.4 Report Generation Testing

PDF reports were generated for a variety of report types and verified for correct content including doctor details, patient details, detection results, and embedded annotated images. Reports with no findings, single findings, and multiple findings were all tested successfully. The PDF download functionality was tested in multiple browsers.

7.5 Data Isolation Testing

Data isolation was rigorously tested by creating two separate doctor accounts and verifying that each account can only retrieve its own reports. Attempts to access another doctor's report by manipulating the report ID in the URL were correctly rejected with 404 Not Found responses, as the backend scopes all report queries by the authenticated doctor's ID.

7.6 Performance Testing

Response times were measured for the most computationally intensive operation, the detection pipeline. On a standard broadband internet connection, the total time from image upload to completed report including the Roboflow API call, bounding box rendering, and PDF generation averaged approximately twelve to eighteen seconds. The Roboflow API call itself accounted for the majority of this time, typically eight to twelve seconds depending on network conditions. Local processing including bounding box rendering and PDF generation typically completed in under three seconds.

8. Deployment

8.1 Deployment Architecture

The production deployment uses three cloud services. MongoDB Atlas provides a managed MongoDB cluster with automatic backups, monitoring, and global distribution. The free tier M0 cluster provides 512 MB of storage, which is sufficient for storing doctor profiles, report metadata, and OTP records. Uploaded images and generated PDF files are stored on the backend server's file system. Render provides a managed Node.js hosting environment for the backend API. The free tier provides one web service with 512 MB of RAM and automatic deployment from a connected GitHub repository. Vercel provides a globally distributed CDN-backed hosting environment for the React frontend. The free tier supports unlimited personal projects with automatic deployment from GitHub.

8.2 Continuous Deployment

The deployment pipeline is configured for continuous deployment from the main branch of the GitHub repository. Any push to the main branch automatically triggers a new build and deployment on both Render and Vercel. This enables rapid iteration and bug fixes without manual deployment steps.

8.3 Environment Configuration

All sensitive configuration values including the MongoDB connection string, JWT secret, Gmail credentials, and Roboflow API keys are stored as environment variables in the respective hosting platforms' secret management systems, never in the source code repository. This follows the twelve-factor app methodology for managing application configuration.

9. Conclusion

9.1 Summary

This project has successfully designed, implemented, tested, and deployed a full-stack AI-powered dental disease detection system tailored for licensed dental practitioners. The system integrates two state-of-the-art deep learning models — a Roboflow 3.0 Object Detection model for OPG radiograph analysis and a Roboflow RF-DETR Instance Segmentation model for plaque and calculus detection — into a secure, user-friendly web portal. The portal provides OTP-verified registration, JWT-based authentication, comprehensive report generation with annotated images and downloadable PDFs, a searchable case history with filtering, and a settings panel with profile editing and dark mode support.

9.2 Contributions

The key technical contributions of this project include the development and training of two dental AI models on custom datasets, the design and implementation of a unified prediction normalization pipeline that handles both object detection and instance segmentation outputs, the development of a server-side Canvas-based image annotation system that renders bounding boxes and segmentation masks on uploaded images, the design and implementation of a structured PDF report generation pipeline, and the development of a complete secure web portal with email-based identity verification.

9.3 Limitations

The system has several limitations that should be acknowledged. The OPG model was trained on a dataset of 1,172 images, which while sufficient for a research prototype, is small compared to the datasets used to train commercial dental AI systems, which may have access to hundreds of thousands of clinical images. The plaque model's mAP@50 of 56.0% reflects the inherent difficulty of the segmentation task and the limited training dataset size of 512 images. Image quality significantly affects model performance; low-resolution, blurry, or poorly lit images may yield inaccurate predictions. The system does not currently support CBCT or bitewing radiograph analysis, which are important in clinical practice. Uploaded images and PDFs are stored on the Render server's ephemeral file system, meaning they may be lost during server restarts; a production deployment would need to use cloud object storage such as Amazon S3 or Cloudinary for persistent file storage.

9.4 Future Work

Several directions for future improvement are identified. First, expanding the training datasets through data collection partnerships with dental clinics and hospitals in Pakistan would significantly improve model accuracy and generalizability. Second, integrating cloud object storage such as Amazon S3 for persistent image and PDF storage would make the system suitable for long-term clinical use. Third, adding support for additional imaging modalities including bitewing radiographs and periapical radiographs would broaden the system's clinical utility. Fourth, developing a mobile application for iOS and Android would make the system more accessible to dentists who prefer to work on mobile devices. Fifth, integrating natural language generation to produce more detailed, contextually appropriate clinical summaries beyond the current template-based approach would enhance the quality of the generated reports. Sixth, implementing federated learning protocols would allow the model to be improved from clinical feedback without exposing sensitive patient data, enabling continuous learning in a privacy-preserving manner.

10.Generative AI Usage Statement

"This project utilized Generative AI tools, including OpenAI's ChatGPT and similar AI-assisted platforms, for limited academic and technical support throughout the development process. AI tools were primarily used for brainstorming project ideas, understanding technical concepts, generating and refining training/testing scripts, debugging programming errors, improving code structure, deployment guidance, and documentation refinement.

During the implementation phase, AI assistance was used to help resolve errors related to machine learning model training, backend/frontend integration, API handling, and deployment issues on Render and Vercel. AI tools were also consulted for improving development workflow, explaining algorithms, refining technical writing, and providing general coding guidance.

All AI-generated suggestions, code snippets, explanations, and documentation were carefully reviewed, modified, tested, and validated by the group members before inclusion in the final system. The students independently performed the final implementation, dataset preparation, model training, testing, evaluation, analysis, result interpretation, deployment, and verification of the project.

The submitted work represents the group's own original effort, and the team takes full responsibility for the accuracy, authenticity, and integrity of the final project."

11 References

Esteva, A., Kuprel, B., Novoa, R. A., Ko, J., Swetter, S. M., Blau, H. M., and Thrun, S. (2017). Dermatologist-level classification of skin cancer with deep neural networks. Nature, 542(7639), 115-118.

Rajpurkar, P., Irvin, J., Zhu, K., Yang, B., Mehta, H., Duan, T., Ding, D., Bagul, A., Langlotz, C., Shpanskaya, K., Lungren, M. P., and Ng, A. Y. (2017). CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. arXiv preprint arXiv:1711.05225.

Schwendicke, F., Golla, T., Dreher, M., and Krois, J. (2019). Convolutional neural networks for dental image diagnostics: A scoping review. Journal of Dentistry, 91, 103226.

Lee, J. H., Kim, D. H., Jeong, S. N., and Choi, S. H. (2018). Detection and diagnosis of dental caries using a deep learning-based convolutional neural network algorithm. Journal of Dentistry, 77, 106-111.

Ekert, T., Krois, J., Meinhold, L., Elhennawy, K., Emara, R., Golla, T., and Schwendicke, F. (2019). Deep learning for the radiographic detection of apical lesions. Journal of Endodontics, 45(7), 917-922.

Redmon, J., Divvala, S., Girshick, R., and Farhadi, A. (2016). You only look once: Unified, real-time object detection. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 779-788.

Lowe, D. G. (2004). Distinctive image features from scale-invariant keypoints. International Journal of Computer Vision, 60(2), 91-110.
Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., and Zagoruyko, S. (2020). End-to-end object detection with transformers. European Conference on Computer Vision, 213-229.
World Health Organization. (2022). Oral health. WHO Fact Sheet. Retrieved from https://www.who.int/news-room/fact-sheets/detail/oral-health
Lin, T. Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollar, P., and Zitnick, C. L. (2014). Microsoft COCO: Common objects in context. European Conference on Computer Vision, 740-755.